UNIT - IV

I/O and File System:

I/O Management:

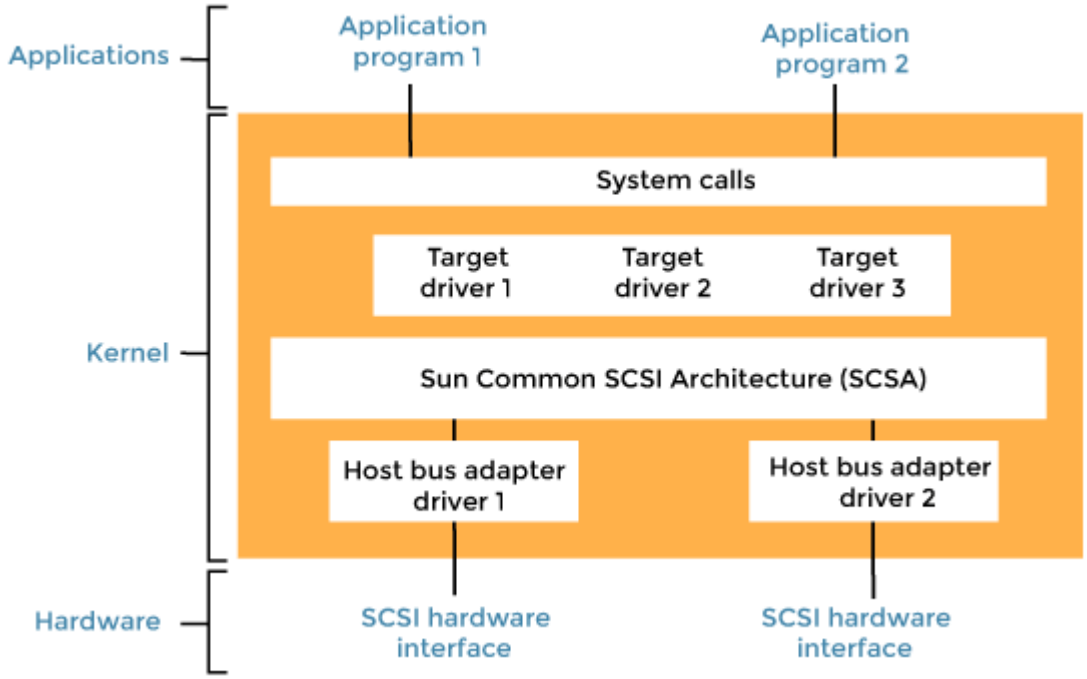
The I/O management subsystem **controls all the input and output of the computer system**. For the enforcement of security, the most important things that the I/O management subsystem does are. managing the transfer of data. enforcing access controls on data while it is being transferred.

Device Driver in Operating System

Device Drivers are important for a computer system to work properly. Without a device driver, the particular hardware fails to work accordingly, which means it fails in doing a particular action for which it has been created.

Drivers are hardware-dependent and operating-system-specific. They usually provide the interrupt handling required for any necessary asynchronous time-dependent hardware interface.

A driver communicates with the device through the computer bus or communications subsystem to which the hardware connects. Once the device sends data back to the driver, the driver may invoke routines in the original calling program. When a calling program invokes a driver's routine, the driver issues the commands to the device.



Device drivers encapsulate device-dependent code and implement a standard interface to contain device-specific register reads or writes. A device driver is generally written by the device's manufacturer and delivered along with the device on a CD-ROM. A device driver performs the following jobs, such as:

- To accept request from the device-independent software above to it.
- Making sure that the request is executed successfully.
- Interact with the device controller to take and give I/O and perform required error handling.

How Device Driver Handles a Request?

Suppose a request comes to read a block N. If the driver is idle when a request arrives, it starts carrying out the request immediately. Otherwise, if the driver is already busy with some other request, it places the new request in the queue of pending requests.

How does Device Driver work?

A device driver is a piece of software that allows your computer's operating system to communicate with a hardware device the driver is written for. Generally, a driver communicates with the device through the ***computer bus***, which connects the device with the computer. Device Drivers depend upon the Operating System's instruction to access the device and performing any particular action. After the action, they also show their reactions by delivering output or message from the hardware device to the Operating system.

Device drivers work within the ***kernel*** layer of the operating system. The kernel is the part of the operating system that directly interacts with the system's physical structure. Instead of accessing a device directly, an operating system loads the device drivers and calls the specific functions in the driver software to execute specific tasks on the device. Each driver contains the device-specific codes required to carry out the actions on the device.

Card reader, controller, modem, network card, sound card, printer, video card, USB devices, RAM, Speakers etc., need Device Drivers to operate. For example, a printer driver tells the printer which format to print after getting instructions from OS. Similarly, A sound card driver is there because the 1's and 0's data of an MP3 file is converted to audio signals, and you enjoy the music.

Types of Device Driver

For almost every device associated with the computer system, a Device Driver exists for the particular hardware. But it can be broadly classified into the following two types, such as:

1. Kernel-mode Device Driver

This Kernel-mode device driver includes some generic hardware that loads with an operating system as part of the OS. These are ***BIOS, motherboard, processor***, and some other hardware that are part of kernel software. These include the minimum system requirement device drivers for each operating system.

- **BIOS:** BIOS (basic input/output system) is the most basic computer driver in existence. It is designed to be the first program that boots when a PC turns on. The BIOS is stored on

memory built into the motherboard and is designed to boot the hardware connected to the PC, including the hard drives, video display output, keyboard and mouse.

- **Motherboard Drivers:** Motherboard drivers are small programs that are read by either Windows or Linux and allow for basic computer functions while inside the operating system. These drivers normally include programs that allow broadband ports, USB ports and I/O ports for the mouse and keyboard.

2. User-mode Device Driver

Other than the devices brought by the kernel for working of the system, the user also brings some devices for use during the using of a system that devices need device drivers to functions those drivers fall under User mode device driver. For example, the user needs any plug and play action that comes under this.

Character Device Vs. Block Device:

- A Character Device is one with which the Driver communicates by sending and receiving single characters (bytes, octets).
- A Block Device is one with which the Driver communicates by sending entire blocks of data.
- Examples for Character Devices: serial ports, parallel ports, sounds cards.
- Examples for Block Devices: hard disks, USB cameras, Disk-On-Key.
- For the user, the type of the Device (block or character) does not matter - you just care that this is a hard disk partition or a sound card.
- Driver programmers take care of this.

File Management:

What is a File ?

A file can be defined as a data structure which stores the sequence of records. Files are stored in a file system, which may exist on a disk or in the main memory. Files can be simple (plain text) or complex (specially-formatted).

The collection of files is known as Directory. The collection of directories at the different levels, is known as **File System.**

Data--→ Files--→ Directories--→File System

Attributes of the File

1.Name

Every file carries a name by which the file is recognized in the file system. One directory cannot have two files with the same name.

2.Identifier

Along with the name, Each File has its own extension which identifies the type of the file. For example, a text file has the extension **.txt,** A video file can have the extension **.mp4.**

3.Type

In a File System, the Files are classified in different types such as video files, audio files, text files, executable files, etc.

4.Location

In the File System, there are several locations on which, the files can be stored. Each file carries its location as its attribute.

5.Size

The Size of the File is one of its most important attribute. By size of the file, we mean the number of bytes acquired by the file in the memory.

6.Protection

The Admin of the computer may want the different protections for the different files. Therefore each file carries its own set of permissions to the different group of Users.

7.Time and Date

Every file carries a time stamp which contains the time and date on which the file is last modified.

Operations on the File

A file is a collection of logically related data that is recorded on the secondary storage in the form of sequence of operations. The content of the files are defined by its creator who is creating the file. The various operations which can be implemented on a file such as read, write, open and close etc. are called file operations. These operations are performed by the user by using the commands provided by the operating system. Some common operations are as follows:

1.Create operation:

This operation is used to create a file in the file system. It is the most widely used operation performed on the file system. To create a new file of a particular type the associated application program calls the file system. This file system allocates space to the file. As the file system knows the format of directory structure, so entry of this new file is made into the appropriate directory.

2. Open operation:

This operation is the common operation performed on the file. Once the file is created, it must be opened before performing the file processing operations. When the user wants to open a file, it provides a file name to open the particular file in the file system. It tells the operating system to invoke the open system call and passes the file name to the file system.

3. Write operation:

This operation is used to write the information into a file. A system call write is issued that specifies the name of the file and the length of the data has to be written to the file. Whenever the file length is increased by specified value and the file pointer is repositioned after the last byte written.

4. Read operation:

This operation reads the contents from a file. A Read pointer is maintained by the OS, pointing to the position up to which the data has been read.

5. Re-position or Seek operation:

The seek system call re-positions the file pointers from the current position to a specific place in the file i.e. forward or backward depending upon the user's requirement. This operation is generally performed with those file management systems that support direct access files.

6. Delete operation:

Deleting the file will not only delete all the data stored inside the file it is also used so that disk space occupied by it is freed. In order to delete the specified file the directory is searched. When the directory entry is located, all the associated file space and the directory entry is released.

7. Truncate operation:

Truncating is simply deleting the file except deleting attributes. The file is not completely deleted although the information stored inside the file gets replaced.

8. Close operation:

When the processing of the file is complete, it should be closed so that all the changes made permanent and all the resources occupied should be released. On closing it deallocates all the internal descriptors that were created when the file was opened.

9. Append operation:

This operation adds data to the end of the file.

10. Rename operation:

This operation is used to rename the existing file.

File Access Methods

Let's look at various ways to access files stored in secondary memory.

Sequential Access

Most of the operating systems access the file sequentially. In other words, we can say that most of the files need to be accessed sequentially by the operating system.

In sequential access, the OS read the file word by word. A pointer is maintained which initially points to the base address of the file. If the user wants to read first word of the file then the pointer provides that word to the user and increases its value by 1 word. This process continues till the end of the file.

Modern word systems do provide the concept of direct access and indexed access but the most used method is sequential access due to the fact that most of the files such as text files, audio files, video files, etc need to be sequentially accessed.

Direct Access

The Direct Access is mostly required in the case of database systems. In most of the cases, we need filtered information from the database. The sequential access can be very slow and inefficient in such cases.

Suppose every block of the storage stores 4 records and we know that the record we needed is stored in 10th block. In that case, the sequential access will not be implemented because it will traverse all the blocks in order to access the needed record.

Direct access will give the required result despite of the fact that the operating system has to perform some complex tasks such as determining the desired block number. However, that is generally implemented in database applications.

Indexed Access

If a file can be sorted on any of the filed then an index can be assigned to a group of certain records. However, A particular record can be accessed by its index. The index is nothing but the address of a record in the file.

In index accessing, searching in a large database became very quick and easy but we need to have some extra space in the memory to store the index value.

Directory Structure

What is a directory?

Directory can be defined as the listing of the related files on the disk. The directory may store some or the entire file attributes.

To get the benefit of different file systems on the different operating systems, A hard disk can be divided into the number of partitions of different sizes. The partitions are also called volumes or mini disks.

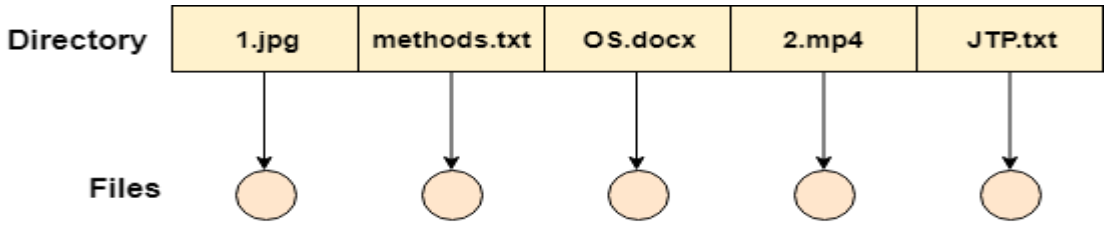
Each partition must have at least one directory in which, all the files of the partition can be listed. A directory entry is maintained for each file in the directory which stores all the information related to that file.

Every Directory supports a number of common operations on the file:

1. File Creation
2. Search for the file
3. File deletion
4. Renaming the file
5. Traversing Files
6. Listing of files

Single Level Directory

The simplest method is to have one big list of all the files on the disk. The entire system will contain only one directory which is supposed to mention all the files present in the file system. The directory contains one entry per each file present on the file system.



Single Level Directory

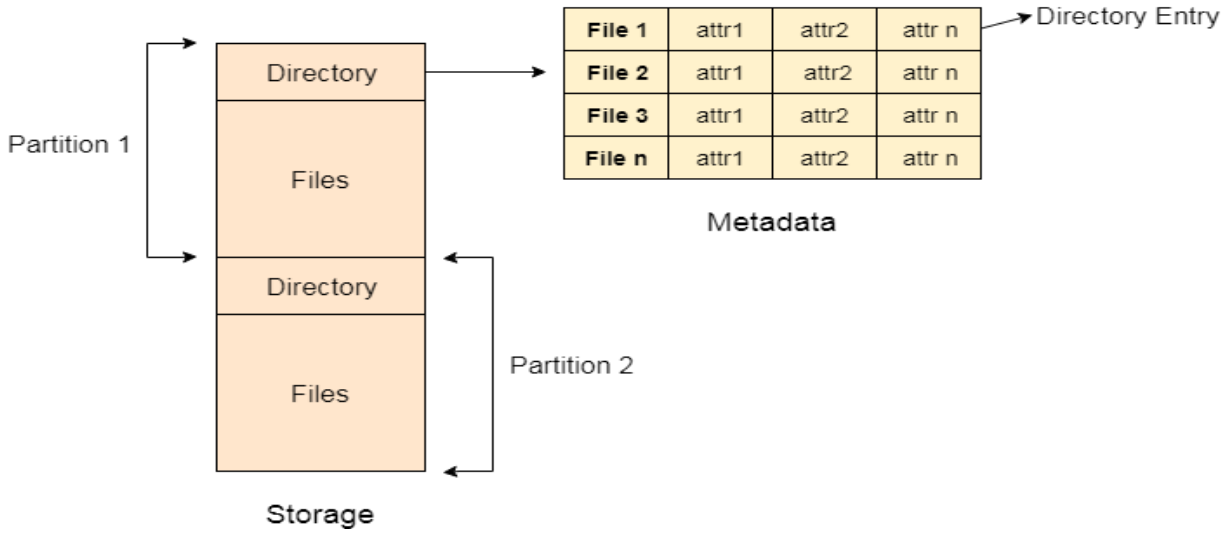
This type of directories can be used for a simple system.

Advantages

1. Implementation is very simple.
2. If the sizes of the files are very small then the searching becomes faster.
3. File creation, searching, deletion is very simple since we have only one directory.

Disadvantages

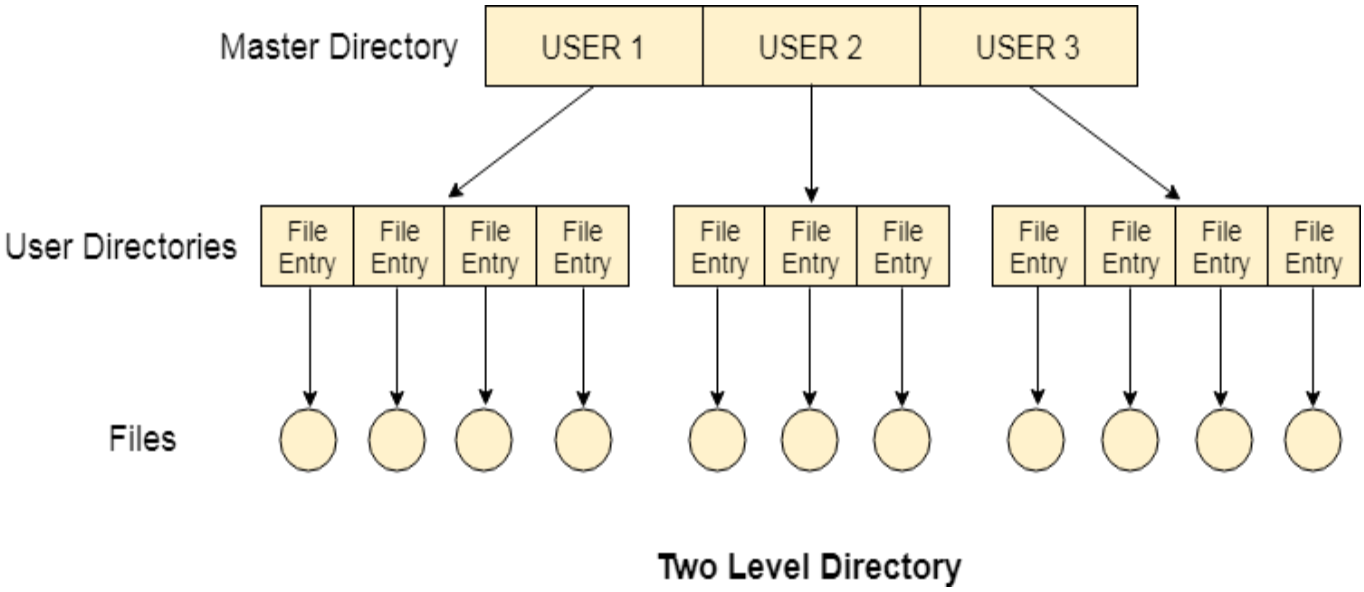
1. We cannot have two files with the same name.
2. The directory may be very big therefore searching for a file may take so much time.
3. Protection cannot be implemented for multiple users.
4. There are no ways to group same kind of files.
5. Choosing the unique name for every file is a bit complex and limits the number of files in the system because most of the Operating System limits the number of characters used to construct the file name.



A directory can be viewed as a file which contains the Meta data of the bunch of files.

Two Level Directory

In two level directory systems, we can create a separate directory for each user. There is one master directory which contains separate directories dedicated to each user. For each user, there is a different directory present at the second level, containing group of user's file. The system doesn't let a user to enter in the other user's directory without permission.



Characteristics of two level directory system

1. Each files has a path name as ***/User-name/directory-name/***
2. Different users can have the same file name.
3. Searching becomes more efficient as only one user's list needs to be traversed.
4. The same kind of files cannot be grouped into a single directory for a particular user.

Every Operating System maintains a variable as **PWD** which contains the present directory name (present user name) so that the searching can be done appropriately.

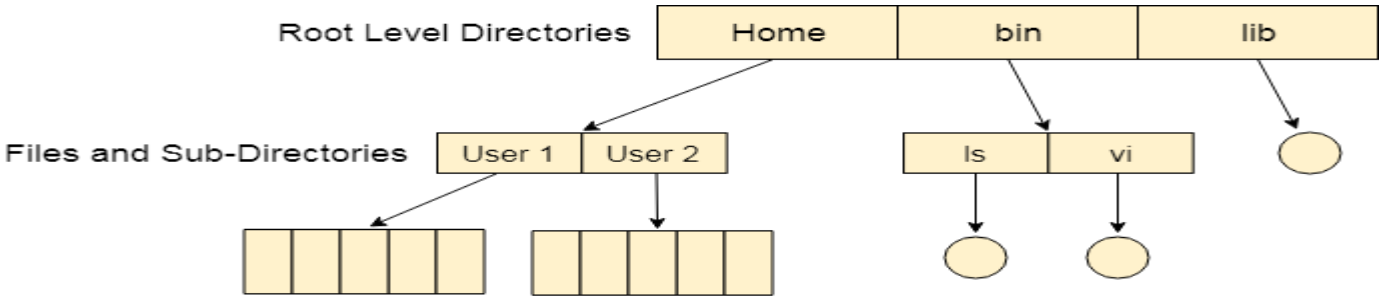
Tree Structured Directory

In Tree structured directory system, any directory entry can either be a file or sub directory. Tree structured directory system overcomes the drawbacks of two level directory system. The similar kind of files can now be grouped in one directory.

Each user has its own directory and it cannot enter in the other user's directory. However, the user has the permission to read the root's data but he cannot write or modify this. Only administrator of the system has the complete access of root directory.

Searching is more efficient in this directory structure. The concept of current working directory is used. A file can be accessed by two types of path, either relative or absolute.

Absolute path is the path of the file with respect to the root directory of the system while relative path is the path with respect to the current working directory of the system. In tree structured directory systems, the user is given the privilege to create the files as well as directories.



The Structured Directory System

Permissions on the file and directory

A tree structured directory system may consist of various levels therefore there is a set of permissions assigned to each file and directory.

The permissions are **R W X** which are regarding reading, writing and the execution of the files or directory. The permissions are assigned to three types of users: owner, group and others.

Acyclic-Graph Structured Directories

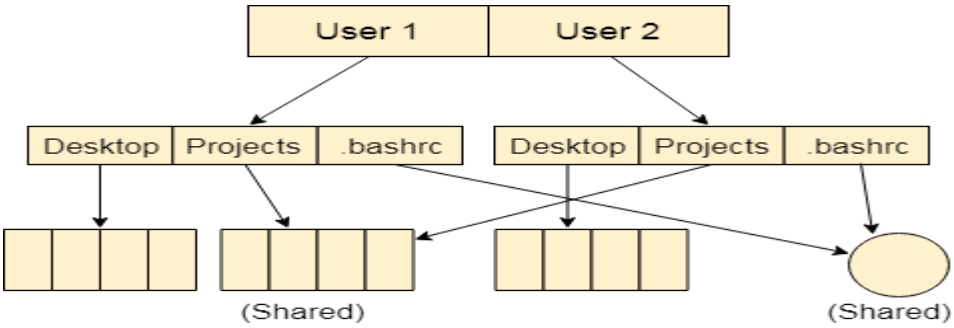
The tree structured directory system doesn't allow the same file to exist in multiple directories therefore sharing is major concern in tree structured directory system. We can provide sharing by making the directory an acyclic graph. In this system, two or more directory entry can point to the same file or sub directory. That file or sub directory is shared between the two directory entries.

These kinds of directory graphs can be made using links or aliases. We can have multiple paths for a same file. Links can either be symbolic (logical) or hard link (physical).

If a file gets deleted in acyclic graph structured directory system, then

1. In the case of soft link, the file just gets deleted and we are left with a dangling pointer.

2. In the case of hard link, the actual file will be deleted only if all the references to it gets deleted.



Acyclic-Graph Structured Directory System

File Systems

File system is the part of the operating system which is responsible for file management. It provides a mechanism to store the data and access to the file contents including data and programs. Some Operating systems treats everything as a file for example Ubuntu.

The File system takes care of the following issues

- **File Structure**

  We have seen various data structures in which the file can be stored. The task of the file system is to maintain an optimal file structure.

- **Recovering Free space**

  Whenever a file gets deleted from the hard disk, there is a free space created in the disk. There can be many such spaces which need to be recovered in order to reallocate them to other files.

- **disk space assignment to the files**

  The major concern about the file is deciding where to store the files on the hard disk. There are various disks scheduling algorithm which will be covered later in this tutorial.

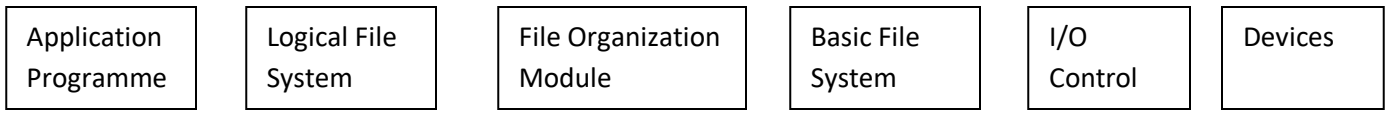
- **tracking data location**

  A File may or may not be stored within only one block. It can be stored in the non contiguous blocks on the disk. We need to keep track of all the blocks on which the part of the files reside.

File System Structure

File System provide efficient access to the disk by allowing data to be stored, located and retrieved in a convenient way. A file System must be able to store the file, locate the file and retrieve the file.

Most of the Operating Systems use layering approach for every task including file systems. Every layer of the file system is responsible for some activities.



- When an application program asks for a file, the first request is directed to the logical file system. The logical file system contains the Meta data of the file and directory structure. If the application program doesn't have the required permissions of the file then this layer will throw an error. Logical file systems also verify the path to the file.
- Generally, files are divided into various logical blocks. Files are to be stored in the hard disk and to be retrieved from the hard disk. Hard disk is divided into various tracks and sectors. Therefore, in order to store and retrieve the files, the logical blocks need to be mapped to physical blocks. This mapping is done by File organization module. It is also responsible for free space management.
- Once File organization module decided which physical block the application program needs, it passes this information to basic file system. The basic file system is responsible for issuing the commands to I/O control in order to fetch those blocks.
- I/O controls contain the codes by using which it can access hard disk. These codes are known as device drivers. I/O controls are also responsible for handling interrupts.

File System Protection:

In computer systems, the objective of the operating system is to keep safe the data of the user from the improper access to the system. Protection can be provided in number of ways. For a single laptop system, we might provide protection by locking the computer in a desk drawer or file cabinet. For multi-user systems, different mechanisms are used for the protection.

Types of Access :
The files which have direct access of the any user have the need of protection. The files which are not accessible to other users doesn’t require any kind of protection. The mechanism of the protection provide the facility of the controlled access by just limiting the types of access to the file. Access can be given or not given to any user depends on several factors, one of which is the type of access required. Several different types of operations can be controlled:

- **Read –** Reading from a file.
- **Write –** Writing or rewriting the file.
- **Execute –** Loading the file and after loading the execution process starts.
- **Append –** Writing the new information to the already existing file, editing must be end at the end of the existing file.
- **Delete –** Deleting the file which is of no use and using its space for the another data.
- **List –** List the name and attributes of the file.

Operations like renaming, editing the existing file, copying; these can also be controlled. There are many protection mechanism. each of them mechanism have different advantages and disadvantages and must be appropriate for the intended application.

Access Control :

There are different methods used by different users to access any file. **The general way of protection is to associate *identity-dependent access* with all the files and directories an list called access-control list (ACL) which specify the names of the users and the types of access associate with each of the user**. The main problem with the access list is their length. If we want to allow everyone to read a file, we must list all the users with the read access. This technique has two undesirable consequences: Constructing such a list may be tedious and unrewarding task, especially if we do not know in advance the list of the users in the system.

Previously, the entry of the any directory is of the fixed size but now it changes to the variable size which results in the complicates space management. These problems can be resolved by use of a condensed version of the access list. To condense the length of the access-control list, many systems recognize three classification of users in connection with each file:

- **Owner –** Owner is the user who has created the file.
- **Group –** A group is a set of members who has similar needs and they are sharing the same file.
- **Universe –** In the system, all other users are under the category called universe.

The most common recent approach is to combine access-control lists with the normal general owner, group, and universe access control scheme. For example: Solaris uses the three categories of access by default but allows access-control lists to be added to specific files and directories when more fine-grained access control is desired.

Other Protection Approaches:
The access to any system is also controlled by the password. If the use of password is random and it is changed often, this may be result in limit the effective access to a file.

The use of passwords has a few disadvantages:

- The number of passwords are very large so it is difficult to remember the large passwords.
- If one password is used for all the files, then once it is discovered, all files are accessible; protection is on all-or-none basis.

File System Security and Integrity:

Video tutorial :

https://www.youtube.com/watch?v=IIKJdqW0uK4

https://www.youtube.com/watch?v=DKb7KhfoZmU

https://www.youtube.com/watch?v=I4re_0qqwXQ